

VesSkel

Vessel Skeletonization and Graph-Based Phenotype Analysis in Retinal Fundus Images

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CLI Throughput: Lung Scans Parallel Thinning

Added .mhd + .raw (MetaImage) file loading via ITK

Benchmark: `vessel` run on the 20 VESSEL12 lung CT masks (loaded via MHD support)

Hardware: 12-core CPU (24 threads) – homogenous CPU Architecture, no thermal throttling

Mode	Wall Time	User CPU	Sys CPU	CPU Util
-j 12 (default)	3:28.94	2376.86s	34.05s	1153%
-j 1 (seq.)	4:53.63	1773.63s	16.75s	609%

Parallel is **28% faster** in wall time (3:29 vs 4:54). Sequential uses **25% less total CPU** (1774s vs 2377s user).

3D Lung Thinning: VesSkel vs other Implementations

Benchmark: benchmark/3d_lung_comparison.py: Lee94 thinning only (preprocessing excluded from timing)

Dataset: 20 VESSEL12 lung CT angiography masks (loaded via MHD)

Stat	vesskel	skimage Lee	VesselVio Lee	Speedup (sk / vv)
TOTAL	335.39s	604.99s	679.38s	1.80x / 2.03x
MEAN	16.77s	30.25s	33.97s	1.80x / 2.03x
MEDIAN	16.95s	28.85s	32.66s	1.70x / 1.93x

Reminder: Prior test on a synthetic volume (vessap test volume) showed modest gains: 1.4x vs skimage, 1.1x vs VesselVio

Napari UX: Conditional Toggles

Problem: Previously all output options were always enableable – confusing when extraction was off -
> would fail/skip silently

Solution: Output widgets auto-disable when their extraction toggle is off:

Output option	Requires	Behavior
Write skeleton	always on	always enabled
Write branch CSV	Extract branches	disabled when off
Write node CSV	Extract nodes	disabled when off
Write radius	Vessel radius	disabled when off
Write summary CSV	Extract summary	disabled when off

Cleanup group separated from Advanced Features – Show preprocessed auto-enables only when closing/hole-fill is active

Branch Coloring by Feature

Before: Branches always colored by tortuosity (hardcoded)

After: Dropdown lets user pick any numeric property – tortuosity, straightness, branch distance, radius stats, etc.

Live recoloring on dropdown change – no re-analysis needed.

Demo time?

Code Architecture & Quality

IO extraction

1. `_load_image`, `_save_skeleton`, `_save_radius`, `_write_csv` moved from `_batch.py` -> new `_io.py`
2. New `save_analysis_outputs()` unifies all output writing in one entry point
3. `write_summary` flag suppresses per-image CSV in batch mode (aggregated summary at top level)

Config refactoring

1. `from_dict` now uses `dataclass` fields introspection – no manual field mapping
2. Schema version bumped 2 -> 3
3. New field `branch_color_property` added

Default changes

1. `extract_summary` now defaults to `True` (was `False`) – CLI and napari always compute summary features

Batch CLI robustness

1. Graceful Ctrl+C shutdown: `KeyboardInterrupt` -> `ex.kill_workers()` -> no hanging workers

Report and Presentation Progress

Report is structurally complete – sections: Intro, Results, Discussion, Methods. Needs final polish pass + formatting tweaks.

Summary & Next Steps

Benchmarks

1. 20-image VESSEL12 lung CT benchmark
2. MHD support is nice for clinical data
3. 1.8x-2.0x speedup vs skimage/VesselVio
4. Parallel scaling does not perform too well

Napari UX

1. Conditional toggles: self-documenting widget
2. Branch coloring dropdown + live recoloring
3. Cleanup group separated from advanced

Code quality

1. Config introspection, schema v3
2. Graceful Ctrl+C in batch CLI
3. Summary on by default

Next steps:

1. Report and Presentation final polish